

# Alexander Elzenaar

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## Research interests

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- Groups related to geometry and topology in low dimensions. Kleinian groups and their deformation theory; other aspects of geometric group theory, conformal geometry, and 3-manifold theory. Applications of techniques from representation theory and algebraic geometry.
- Exemplar MSC codes: 20F65, 20H10, 22E40, 30F40, 30F60, 32G15, 51F15, 57K31, 57K32.

## Tertiary education

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| <b>School of Mathematics, Monash University</b><br><i>Doctoral student, geometry and topology</i><br>Advisor: Prof. Jessica Purcell.<br>Expected completion: 2027.                                                                                                                                                                                                                   | <b>Melbourne, Australia</b><br>2024–ongoing |
| <b>Dept. of Mathematics, The University of Auckland</b><br><i>MSc with First Class Honours in mathematics</i><br>Advisors: Dist. Prof. Gaven Martin (NZ Inst. Adv. Study., Massey Uni.) and Dr. Jeroen Schillewaert.<br>Thesis: <i>Deformation spaces of Kleinian groups</i> ( <a href="https://aelzenaar.github.io/msc_thesis.pdf">https://aelzenaar.github.io/msc_thesis.pdf</a> ) | <b>Auckland, NZ</b><br>2021–2022            |
| <b>Dept. of Mathematics, The University of Auckland</b><br><i>BSc (Hons) with First Class Honours in mathematics</i><br>Advisor: Dr. Jeroen Schillewaert.<br>Dissertation: <i>Toric varieties</i> ( <a href="https://aelzenaar.github.io/hons/dissertation.pdf">https://aelzenaar.github.io/hons/dissertation.pdf</a> )                                                              | <b>Auckland, NZ</b><br>2020                 |
| <b>School of Cultures, Languages and Linguistics, The University of Auckland</b><br><i>Certificate of Languages</i><br>Russian and Ancient Egyptian                                                                                                                                                                                                                                  | <b>Auckland, NZ</b><br>awarded 2020         |
| <b>The University of Auckland</b><br><i>BSc, major in mathematics</i>                                                                                                                                                                                                                                                                                                                | <b>Auckland, NZ</b><br>2017–2019            |
| <b>The University of Canterbury</b><br><i>STAR programme</i><br>Distance learning first year university mathematics for secondary school students                                                                                                                                                                                                                                    | <b>Christchurch, NZ</b><br>2016             |

## Employment history (research and non-academic)

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| <b>Ministry of Business, Innovation, and Employment</b><br><i>Research and data analyst, Evidence &amp; Insights branch</i><br>Mathematical modelling and statistics to support ministerial offices and policy evaluators. Contributions to, and peer review of, public research output.                                                                        | <b>Wellington, NZ</b><br>2023–2024   |
| <b>Max-Planck-Inst. für Math. in den Naturwissenschaften</b><br><i>Doctoral student</i><br>Non-linear algebra group. Withdrew for personal reasons.                                                                                                                                                                                                             | <b>Leipzig, Germany</b><br>2022–2023 |
| <b>Dept. of Mathematics, The University of Auckland</b><br><i>Research assistant</i> <ul style="list-style-type: none"><li>- Construction and symmetry properties of spherical <math>(t, t)</math>-designs (funded by Dr. Shayne Waldron)</li><li>- Teaching of semester-long graduate seminar on Kleinian groups (funded by Dr. Jeroen Schillewaert)</li></ul> | <b>Auckland, NZ</b><br>2020–2021     |

## Employment history (teaching)

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**School of Mathematics, Monash University**

**Melbourne, Australia**

*Sessional teaching associate*

2025–2026

- Tutoring and marking MAT 1030 (linear algebra service course), sem. 1, 2026
- Tutoring and marking MAT 1020 (calculus service course), sem. 1, 2025

**Dept. of Mathematics, The University of Auckland**

**Auckland, NZ**

*Graduate teaching assistant*

2019–2022

- Marking MATHS 208 (multivariate calculus service course), summer sem., 2022
- Tutoring & marking MATHS 332 (real analysis); tutoring MATHS 190 (misc. topics inc. cryptography, infinities, tilings, fractals), sem. 2, 2021
- Tutoring & marking MATHS 250 (calculus & linear algebra II); tutoring COMPSCI 225 (discrete mathematics for CS); marking MATHS 254 (number theory; groups; metric spaces), sem. 1, 2021
- Tutoring & marking MATHS 190 (misc. topics inc. cryptography, infinities, tilings, fractals); tutoring MATHS 120 (linear algebra); marking MATHS 340 (real & complex calculus), sem. 2, 2020
- Tutoring MATHS 162 (computational mathematics & combinatorics); assistance room tutor; marking MATHS 120 (linear algebra), MATHS 130 (calculus), COMPSCI 120 (mathematics for CS), sem. 1, 2020
- Marking MATHS 253 (calculus & linear algebra III), sem. 2, 2019

## Academic works

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### Preprints and submitted.....

- [1] A. ELZENAAR, On rank two Kleinian groups with three parabolics, 2026. arXiv 2607.07424.
- [2] A. ELZENAAR, Expansion joints in hyperbolic manifolds, 2025. arXiv 2512.00879.
- [3] A. ELZENAAR, From disc patterns in the plane to character varieties of knot groups, 2025. arXiv 2503.13829.
- [4] A. ELZENAAR, Peripheral subgroups of Kleinian groups, 2025. arXiv 2508.00297.
- [5] A. ELZENAAR, Changing topological type of compression bodies through cone manifolds, 2024. arXiv 2411.17940.
- [6] A. ELZENAAR, J. GONG, G. J. MARTIN, and J. SCHILLEWAERT, Bounding deformation spaces of Kleinian groups with two generators, 2024, to appear in *Annales de l'Institut Fourier*. arXiv 2405.15970.
- [7] A. ELZENAAR, G. J. MARTIN, and J. SCHILLEWAERT, On thin Heckoid and generalised triangle groups in  $\mathrm{PSL}(2, \mathbb{C})$ , NUS/IMS Scientific Reports: Computational Aspects of Thin Groups, 03 Jun 2024–14 Jun 2024, 2024. arXiv 2409.04438.

### Published.....

- [8] A. ELZENAAR, G. J. MARTIN, and J. SCHILLEWAERT, The combinatorics of Farey words and their traces, *Groups, Geometry, and Dynamics* **20** no. 1 (2026), 215–256. MR 5032293. Zbl 8166380. arXiv 2204.08076. <https://doi.org/10.4171/GGD/832>.
- [9] A. ELZENAAR and S. WALDRON, Putatively optimal projective spherical designs with little apparent symmetry, *Journal of Combinatorial Designs* **33** no. 6 (2025), 222–234. MR 4895104. Zbl 1562.05025. arXiv 2405.19353. <https://doi.org/10.1002/jcd.21979>.
- [10] A. ELZENAAR, G. J. MARTIN, and J. SCHILLEWAERT, Concrete one complex dimensional moduli spaces of hyperbolic manifolds and orbifolds, in *2021–22 MATRIX Annals* (D. R. WOOD, J. DE GIER, and C. E. PRAEGER, eds.), Springer, 2024, pp. 31–74. MR 4807267. Zbl 1520.30060. arXiv 2204.11422.

- [11] A. ELZENAAR, G. J. MARTIN, and J. SCHILLEWAERT, Approximations of the Riley slice, *Expositiones Mathematicae* **41** no. 1 (2023), 20–54. MR 4557273. Zbl 1520.30060. arXiv 2111.03230. <https://doi.org/10.1016/j.exmath.2022.12.002>.

## Awards and scholarships

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- Program associate fellowship, SLMath, Topological and Geometric Structures in Low Dimensions Program (Jan. to Mar. 2026)
- CARMA-MATRIX Maths Art Poster Competition, first prize in the outreach category for *Fractals that parameterise hyperbolic handlebodies* (2025)
- Research Training Program Scholarship and Stipend (Australian Govt.) (tenure 2024–2027)
- Clay Mathematics Institute Early Career Researcher Support for attendance of NZMRI Summer School on Groups and Dynamics, Nelson (2023)
- Kalman Summer Scholarship for attendance of NZMRI summer meeting on number theory, Akaroa (2022)
- Uni. of Auckland Dept. of Mathematics Student Research Conference prize (2021)
- Kalman Summer Scholarship for attendance of NZMRI summer meeting, Napier (2021)
- Uni. of Auckland Postgraduate Honours Scholarship (tenure 2020)
- Uni. of Auckland Summer Research Scholarship for *Numerical construction of spherical  $(t, t)$ -designs* with Dr. Shayne Waldron (2019)
- Uni. of Auckland Faculty of Arts Dean's List (2017)
- NZQA Outstanding Scholar Award, limited to top 40–60 secondary school students in New Zealand, including an Outstanding Scholarship in Classical Studies (one of 13 awarded nationally) (2016)
- Royal Society of New Zealand scholarship to attend XVI Summer Research School in Mathematics and Informatics, Blagoevgrad, Bulgaria (2016)

## Selected talks

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- “Modern and classical geometry of circles: AKA, What does algebra have to do with knot theory?”, Oberseminar Deformationsquantisierung und Geometrie (Uni. Würzburg), 16 July 2026.
- “Towards a structure theory for rank two Kleinian groups”, Low-Dimensional Topology workshop (Mathematisches Forschungsinstitut Oberwolfach), 9 July 2026.
- “Quantifying shape and position of discreteness loci in  $\mathrm{PSL}(2, \mathbb{C})$ ”, Séminaire (Laboratoire de Mathématiques de Reims), 23 June 2026.
- “Hands-on cone deformations of hyperbolic manifolds”, Geometry, Dynamics, and Zeta Functions (Radboud Uni., Nijmegen), 18 June 2026.
- “Quantifying shape and position of discreteness loci in  $\mathrm{PSL}(2, \mathbb{C})$ ”, Postgraduate seminar, Topological and Geometric Structures in Low Dimensions (SLMath, Berkeley), 13 March 2026.
- “Generalised Ford domains arising from peripheral geometry”, 69th Annual Meeting of the Australian Mathematical Society (La Trobe Uni.), 9 December 2025.
- “What do knots have to do with algebra?”, 2025 NZMS Colloquium (Uni. of Waikato), 27 November 2025.
- “Discontinuous subgroups of  $\mathrm{Aut}(\mathbb{S}^2)$  come in real-algebraic families with stable combinatorics”, 9th Australian Algebra Conference (La Trobe Uni.), 18 November 2025.
- “Cone manifolds and compression bodies”, Topology Seminar (Monash Uni.) 22 October 2025.
- “Deformations of 3-orbifold holonomy groups and applications”, Early Career Showcase in Low-Dimensional Topology, Joint Meeting of the NZMS, AustMS, and AMS (Uni. of Auckland), 11 December 2024.

- “Combinatorial structures in trace polynomials of function groups”, 8th Australian Algebra Conference (ANU, Canberra), 28 November 2024.
- “Two-bridge knots, genus two surfaces, and discrete groups with two generators”, Hodgsonfest: Geometry and topology in low dimensions (Uni. Melbourne) 12 November 2024.
- “Is  $\mathrm{PSL}(2, \mathbb{C})$  discrete?”, Topology Seminar (Monash Uni.) 24 July 2024.
- “The dynamic in the static: Manifolds, braids, and classical number theory”, Regiomontanus PhD Seminar (Uni. Leipzig), 10 May 2023.
- “What is a Kleinian group?”, Australian Postgraduate Algebra Colloquium, 21 September 2022.
- “Pictures of hyperbolic spaces”, Discrete Mathematics and Geometry Seminar (TU Berlin), 4 May 2022.
- “Strange circles: The Riley slice of quasi-Fuchsian space”, Seminar on Nonlinear Algebra (MPI MiS), 27 April 2022.
- “Approximating the Riley slice exterior”, Matrix Inst. workshop on Groups and Geometries, 2 December 2021.
- “Real varieties of spherical designs”, Algebra and Combinatorics Seminar (Uni. of Auckland), 1 April 2021.

## Professional service

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- Refereeing for scholarly publications.
- Reviewing for zbMATH Open.

## Selected events organised.....

- Minicourse on knot theory and geometry (Uni. Auckland), including guest lectures by Lavender Marshall and Josh Lehman (July 2023).
- Reproducibility in Computer Algebra event (MPI-MiS/TU Berlin), co-organised with Christiane Görgen and Lars Kastner (August 2022).
- A Day of Geometry and Lorentzian Polynomials (MPI-MiS), single day of short informal talks in preparation for Göran Gustafsson Symposium at the Institut Mittag-Leffler (May 2022).
- Seminar on Kleinian groups (Uni. Auckland), weekly lectures covering approximately the first half of *Kleinian Groups* by B. Maskit along with some sections of Thurston’s lecture notes and the book *Hyperbolic Manifolds and Discrete Groups* by M. Kapovich. Funded by Jeroen Schillewaert through a Research Assistantship (2021).
- Reading group on algebraic geometry (Uni. Auckland), jointly organised with Oliver Li (2020–2021).

## Professional affiliations

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- New Zealand Mathematical Society, Student member
- Australian Mathematical Society, Student member

## References

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Teaching and research reference letters available on request.

[July 21, 2026]